

2)Implementation of CANDIDATE ELIMINATION ALGORITHM:

CODE:

```
# Dataset: [Programming Language, CGPA, Internship, Placement]
```

```
data = [  
    ['Python', 'High', 'Internship', 'Yes'],  
    ['Python', 'High', 'No Internship', 'Yes'],  
    ['Java', 'Low', 'Internship', 'No'],  
    ['Python', 'High', 'Internship', 'Yes']  
]
```

```
S = ['0'] * (len(data[0]) - 1)
```

```
G = ['?'] * (len(data[0]) - 1)
```

```
for row in data:
```

```
    x, y = row[:-1], row[-1]
```

```
    if y == 'Yes':
```

```
        if S[0] == '0':
```

```
            S = x.copy()
```

```
        else:
```

```
            for i in range(len(S)):
```

```
                if S[i] != x[i]:
```

```
                    S[i] = '?'
```

```
    else:
```

```
        for i in range(len(S)):
```

```
            if S[i] != '?' and S[i] != x[i]:
```

```
                G[i] = S[i]
```

```
print("NAME:Balakumaran,Reg.No:192412179")
```

```
print("Specific Hypothesis (S):", S)
```

```
print("General Hypothesis (G):", G)
```

OUTPUT:

```
# Dataset: [Programming Language, CGPA, Internship, Placement]
data = [
    ['Python', 'High', 'Internship', 'Yes'],
    ['Python', 'High', 'No Internship', 'Yes'],
    ['Java', 'Low', 'Internship', 'No'],
    ['Python', 'High', 'Internship', 'Yes']
]

S = ['0'] * (len(data[0]) - 1)
G = ['?'] * (len(data[0]) - 1)

for row in data:
    x, y = row[:-1], row[-1]

    if y == 'Yes':
        if S[0] == '0':
            S = x.copy()
        else:
            for i in range(len(S)):
                if S[i] != x[i]:
                    S[i] = '?'
    else:
        for i in range(len(S)):
            if S[i] != '?' and S[i] != x[i]:
                G[i] = S[i]

print("NAME:Balakumaran,Reg.No:192412179")
print("Specific Hypothesis (S):", S)
print("General Hypothesis (G):", G)

... NAME:Balakumaran,Reg.No:192412179
Specific Hypothesis (S): ['Python', 'High', '?']
General Hypothesis (G): ['Python', 'High', '?']
```